WORKSHEET 8.3

Rate In, Rate Out

Unit 8: Applications of Integration

Key ideas

- The integral of a rate is an amount: $\int_a^b R(t)\,dt$ is the amount that entered from $t = a$ to $t = b$.
- Amount at time t = starting amount $+ \int_0^t (\text{in} - \text{out})\,dx$.
- The amount is increasing when the rate in exceeds the rate out.
- A greatest or least amount occurs at an endpoint or where rate in = rate out. Compare the amount at every candidate.

Worked examples

Example 1 · The amount

A tank holds 50 litres. Water enters at 12 litres per minute and leaves at $2t$ litres per minute, for $0 \le t \le 10$. How much is in the tank at $t = 10$?

Solution. $50 + \int_0^{10} (12 - 2t)\,dt = 50 + 120 - 100$.

70 litres.

Example 2 · The greatest amount

When does that tank hold the most water, and how much?

Solution. $12 = 2t$ at $t = 6$; the net rate changes from positive to negative there. Amount: $50 + 72 - 36$.

86 litres, at $t = 6$.

Example 3 · Increasing or decreasing

Is the amount of water in that tank increasing or decreasing at $t = 8$?

Solution. Net rate $12 - 16 = -4$ litres per minute.

Decreasing, because the rate out exceeds the rate in.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

Practice

A calculator may be used where a decimal answer is requested. Give decimals to three places.

CORE

1. A tank holds 40 litres. Liquid enters at 8 litres per minute and leaves at 5 litres per minute. How much is in the tank after 10 minutes?

2. $R(t)$ is the rate, in litres per hour, at which oil enters a tank. Explain the meaning of $\int_0^6 R(t)\,dt$.

3. Sand enters a bin at $6t$ kg per minute and leaves at t^2 kg per minute, for $0 \le t \le 10$. On what interval is the amount of sand increasing?

4. The bin in Question 3 holds 10 kg at $t = 0$. How much does it hold at $t = 6$?

EXAM LEVEL

5. Coffee beans are loaded into a hopper at $L(t) = 18e^{-0.15t}$ kg per minute and used at $U(t) = 4 + 0.5t$ kg per minute, for $0 \le t \le 10$. The hopper holds 25 kg at $t = 0$. How many kilograms are loaded during the 10 minutes?

6. Is the amount of beans in the hopper increasing or decreasing at $t = 6$? Give a reason.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

7. Find the amount of beans in the hopper at $t = 10$.

8. Find the time at which the hopper holds the most beans, and that amount. Justify your answer.

Question 9 · Free-response practice

Fuel enters a tank at the rate $F(t) = 8 + 4\sin\left(\frac{t}{2}\right)$ litres per minute and is drawn off at the rate $D(t) = 0.3t^2$ litres per minute, for $0 \leq t \leq 8$. The tank holds 30 litres at $t = 0$.

(a) How many litres of fuel enter the tank for $0 \leq t \leq 8$?

(b) Find the rate at which the amount of fuel in the tank is changing at $t = 5$. Include units.

(c) Find the greatest amount of fuel in the tank for $0 \leq t \leq 8$. Justify your answer.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.

Worked solutions

1. $40 + 10(8 - 5) = 70$ litres.
2. The number of litres of oil that enter the tank during the first 6 hours.
3. $6t > t^2$ for $0 < t < 6$.
4. $10 + \displaystyle\int_0^6 (6t - t^2)\,dt = 10 + 108 - 72 = 46$ kg.
5. $\displaystyle\int_0^{10} L(t)\,dt \approx 93.224$ kg.
6. $L(6) \approx 7.318$ and $U(6) = 7$. Increasing, because $L(6) > U(6)$.
7. $25 + \displaystyle\int_0^{10} (L(t) - U(t))\,dt \approx 53.224$ kg.
8. $L(t) = U(t)$ at $t \approx 6.201$. Amounts: 25 at $t = 0$, about 63.244 at $t \approx 6.201$, and 53.224 at $t = 10$. The most is about 63.244 kg.
9. (a) $\displaystyle\int_0^8 F(t)\,dt \approx 77.229$ litres.
   (b) $F(5) - D(5) \approx 2.894$ litres per minute.
   (c) $F(t) = D(t)$ only at $t \approx 5.587$. Amounts: 30 at $t = 0$, $30 + \displaystyle\int_0^{5.587} \big(F(t) - D(t)\big)\,dt \approx 72.777$ at $t \approx 5.587$, and about 56.029 at $t = 8$. The greatest amount is about 72.777 litres.

AP© is a registered trademark of the College Board, which was not involved in the production of, and does not endorse, this material. © Integration Academy.